

Monthly Intelligence Report

Edition 003 — The Dublin–Cork Grid Corridor

Where the energy transition is outrunning the grid. edition · June 2026 · SSI v4.0.2

CONTENTS

A Executive Summary	E European Context Card
B Cyber & Economic Exposure Monitor	F SSI Enhanced Neural Network
C Data Refresh & Changelog	G Looking Ahead
D Deep-Dive: The Dublin–Cork Grid Corridor	

SECTION A

Executive Summary

SUBSTATIONS SCORED

994
319 HV · 675 MV

GRID LINES MAPPED

2,306
transmission + distribution

MC SIMULATIONS

9.9 M
10,000 per substation

PUBLIC DATA SOURCES

30+
95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 10,000+ source data points per run, executes 9.9 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 223 million arithmetic operations — producing 59,243 output data points with full uncertainty quantification across 994 substations and 2,306 power lines.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources, making the methodology fully auditable and reproducible. Initially covering Ireland's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Bruckana 38kV Substation

Leinster, Laois · 38 kV

0.278

R_FINAL

Medium

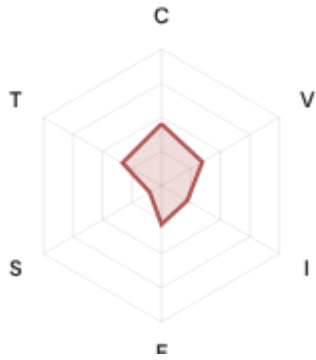
CLASSIFICATION

0.042

CI WIDTH

19th

FLEET PERCENTILE



COMPONENTS

C Continuity: 0.451 / 0.30 (150%)

I Infrastructure: 0.219 / 0.25 (88%)

S Saturation: 0.100 / 0.20 (50%)

V Voltage: 0.347 / 0.10 (347%)

E Economic: 0.286 / 0.10 (286%)

T Transition: 0.331 / 0.05 (662%)

MODIFIERS

R3 Consequence: 0.959 ↓ dampens

R4 Graph Criticality: 0.999 → neutral

R6a Restoration: 0.970 ↓ dampens

R6b Seismic: 1.105 ↑ amplifies

R7 Digital Readiness: 0.933 ↓ dampens

This month's spotlight — automatically selected for June 2026 — is **Bruckana 38kV Substation** in Leinster, Laois. With an R_final of 0.278 (Medium band, 19th fleet percentile), its risk profile is dominated by the T (Transition) component at 662% of its weight ceiling, followed by V (Voltage) at 347%. Modifiers collectively shift R_base by 4.2%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Ireland's National Energy & Climate Plan (NECP) and Climate Action Plan digitalisation priorities require.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

33

High/Critical + R7 < median

AVG R7 MODIFIER

0.9797

Range [0.99, 1.05]

HIGH-CONSEQUENCE SUBS

358

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



● Low ● Medium ● High ● Critical

Dashed lines = fleet medians

The cyber-physical matrix identifies **33 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 0.9800$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Dublin (33)**. Across the full fleet, 0 (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.



Capital-Intensive / High-Consequence

Est. VoLL: €15-30/kWh

307 substations (30.9%) High/Critical: **72** (23.5%) Avg R: **0.448** Avg E: **0.362** / 0.10



Industrial / Medium-Large Enterprise

Est. VoLL: €8-15/kWh

118 substations (11.9%) High/Critical: **9** (7.6%) Avg R: **0.408** Avg E: **0.354** / 0.10



Commercial / Mixed

Est. VoLL: €3–8/kWh

81 substations (8.1%) High/Critical: 4 (4.9%) Avg R: 0.389 Avg E: 0.352 / 0.10



Light-Rural / Agricultural

Est. VoLL: €1–3/kWh

488 substations (49.1%) High/Critical: 3 (0.6%) Avg R: 0.306 Avg E: 0.317 / 0.10



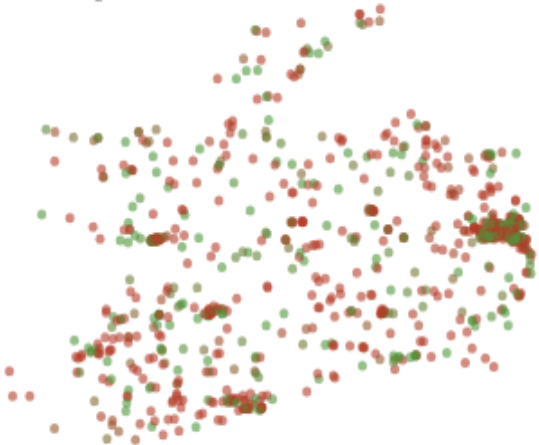
Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **72 substations** combine capital-intensive economic fabric ($R3 \geq 1.14$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **Dublin (222), Cork (44), Limerick (17)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 7.5%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 County Digital Readiness Ranking

Counties ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_{median} to identify counties combining digital exposure with high grid stress.

R7 Digital Readiness
Low High



WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	Leinster ▲ high R		0.9786
2	Munster		0.9791
3	Ulster		0.9799
4	Connacht		0.9843

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Leinster** has the lowest average R7 (0.9786), while **Connacht** leads at 0.9843 — a spread of 0.0057 across the R7 range. **1 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7’s dynamic range and incorporate operational technology (OT) security indicators.

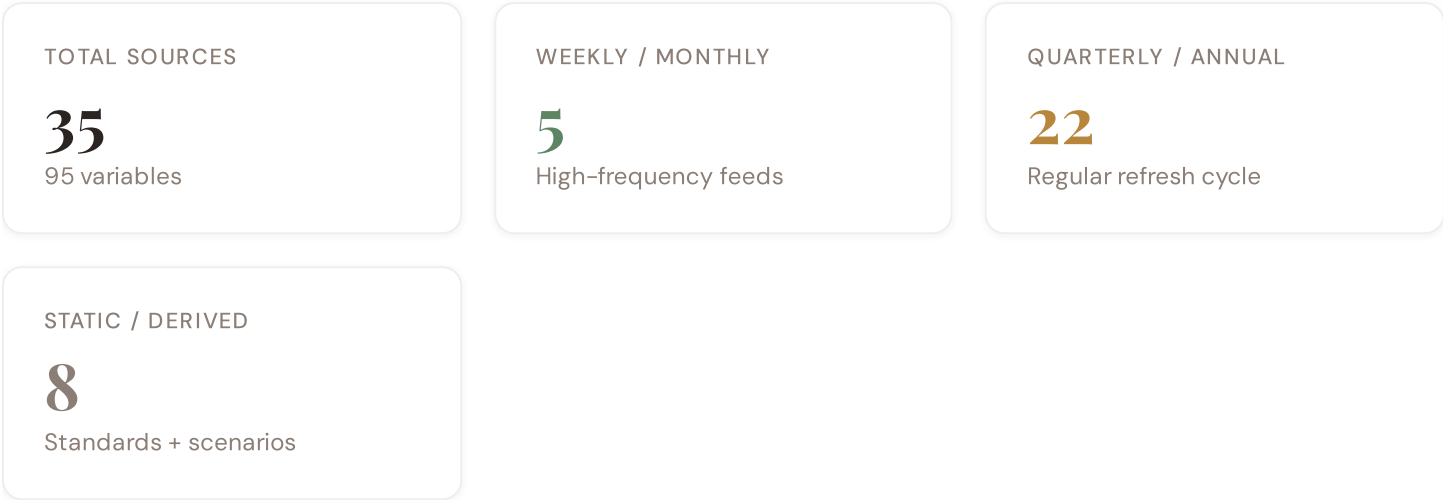
B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 0.9797, median = 0.9800; 0 substations classified HIGH cyber-exposure; 33 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **35 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI’s credibility.



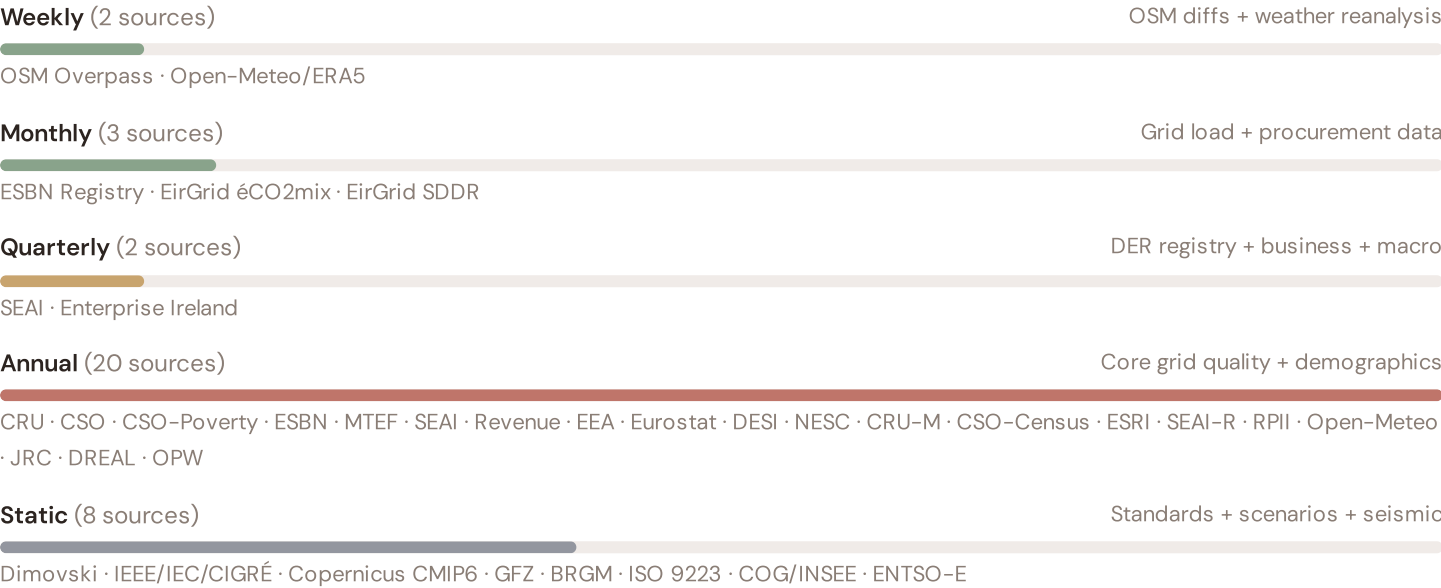
Data Source Registry — June 2026

OSM	OSM Power Infrastructure	WEEKLY	Node/edge	3 vars
EirGrid-N	EirGrid TDP (Transmission Development Plan)	MONTHLY	TSO zone	2 vars
DREAL	EPA (Environmental Protection Agency)	CONTINUOUS	County	1 var
ODRE	ESBN Open Data (ESB Networks)	QUARTERLY	County	5 vars
SEAI	SEAI EV Data	QUARTERLY	County	1 var
BPI	Enterprise Ireland Innovation	QUARTERLY	County	1 var
CRU	CRU (Commission for Regulation of Utilities)	ANNUAL	County	8 vars
CSO	CSO (Central Statistics Office)	ANNUAL	County	8 vars
SEAI	SEAI (Sustainable Energy Authority Ireland)	ANNUAL	County	4 vars
JRC	JRC DSO Observatory	ANNUAL	DSO-level	2 vars
EEA	EEA Air Quality e-Reporting	ANNUAL	~1 km + station	3 vars
EURO	Eurostat Energy Statistics	ANNUAL	NUTS2	3 vars
DESI	DESI Digital Economy Index	ANNUAL	EU Regional	2 vars
MTEF	Department of Environment, Climate and Communications	ANNUAL	Province	2 vars
CEREMA	Cerema (Centre d'études risques)	ANNUAL	County	1 var
Revenue	Revenue Commissioners Statistics	ANNUAL	County	1 var
RPII	RPII Nuclear Safety	ANNUAL	County	1 var
CSO-Poverty	CSO Poverty Statistics	ANNUAL	County	2 vars
ESBN	ESB Networks Open Data	ANNUAL	County	2 vars
NESC	NESC (National Economic and Social Council)	ANNUAL	County	2 vars
CRU-M	CRU Annual Report	ANNUAL	DSO-level	2 vars
CSO-Census	CSO Census Geography	ANNUAL	County	1 var

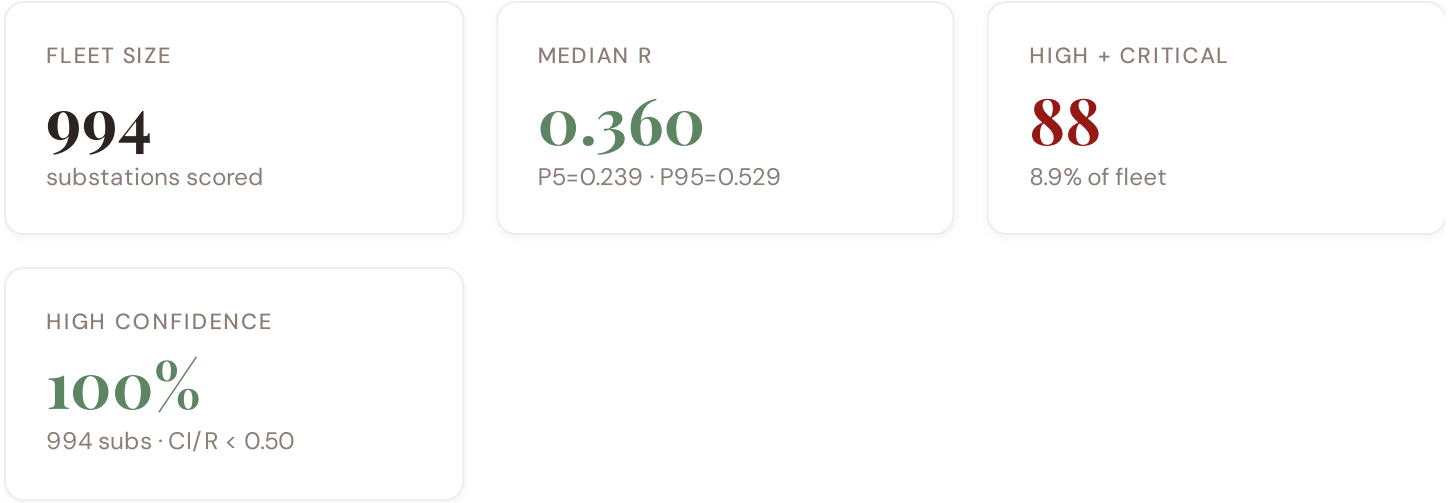
ESRI	ESRI (Economic and Social Research Institute)	ANNUAL	Province	1 var
SEAI-R	SEAI Regions	ANNUAL	Province	1 var
BRGM	GSI (Geological Survey Ireland)	STATIC	County	3 vars
D10	Copernicus CDS / ERA5	STATIC	0.25° (~25 km)	4 vars
DIM	Dimovski et al. (2025)	STATIC	Municipal	3 vars
IEEE-1	IEEE C57.91 / IEC 60076	STATIC	Asset-level	16 vars
GFZ	GFZ German Research Centre	STATIC	~5 km	2 vars
COG	CSO County Registry	STATIC	County	1 var
ISO-9223	ISO 9223 Corrosion	DERIVED	County	1 var
EirGrid	EirGrid Data Portal	HOURLY	Bidding zone	3 vars
ENTSE	ENTSO-E Transparency	HOURLY	Bidding zone	2 vars
MF-0M	Open-Meteo Historical Weather	HOURLY	~1 km	3 vars
D-0M	Open-Meteo / ERA5	HOURLY	~1 km	1 var

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS



C.2 Fleet Score Snapshot



No primary data sources have been updated since the v4.0.2 launch baseline. All **994 substation scores remain unchanged** this edition. The fleet median R of 0.360 (mean 0.369) reflects a distribution skewed toward the lower bands, with 6.7% in Low and 84.4% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

17 changes tracked across PO, P1, and P2 releases

F1	NEW	New T component — Energy Transition Exposure (T1)	\$2, \$4
F2	NEW	New R6a — Restoration Speed Modifier (CAIDI-based)	\$5
F3	ENHANCED	R3 enhanced — Energy Poverty Vulnerability (V_socio)	\$5
F4	ENHANCED	R4 enhanced — Graph-theoretic betweenness + bridge detection	\$5
F5	ENHANCED	R2 enhanced — Climate Trajectory (CMIP6 SSP2-4.5)	\$5
F6	ENHANCED	R7 Digital Readiness — County-level DESI model	\$5
L1	NEW	New I5, I7–I9 metrics — thermal, load, corrosion, hydrogeo	\$2, \$8
L2	ENHANCED	E2 Innovation Enrichment — HRST + startup density	\$6
L3	ENHANCED	V_socio Fiscal Enrichment — DGFIIP + ANCT + energy price	\$5
L4	ENHANCED	R3 Demographic Shift Amplifier — INSEE net migration	\$5
L5	DATA	95/95 variables operational (100%). 35 data sources total.	\$8, \$12
G1	DATA	d02 ESNB upgraded to quarterly registry — County-level DER registry	\$12

G2	DATA	d05 BRGM upgraded to live API — County-level geological data	\$12
G3	DATA	d06 OSM upgraded to Overpass API — 994 real substations	\$12
G4	NEW	R6b Network Topology modifier — centrality + ring analysis from OSM graph	\$5
G5	NEW	Network & Topology layer (H): 7 variables — EirGrid TDP (Transmission Development Plan), ring analysis	\$12
G6	NEW	Output Scores layer (I): 7 variables — risk_score, ETTC, stationary probs	\$12

SECTION D — MONTHLY DEEP-DIVE

The Dublin–Cork Grid Corridor: Where Wind Integration Meets Grid Modernisation Pressure

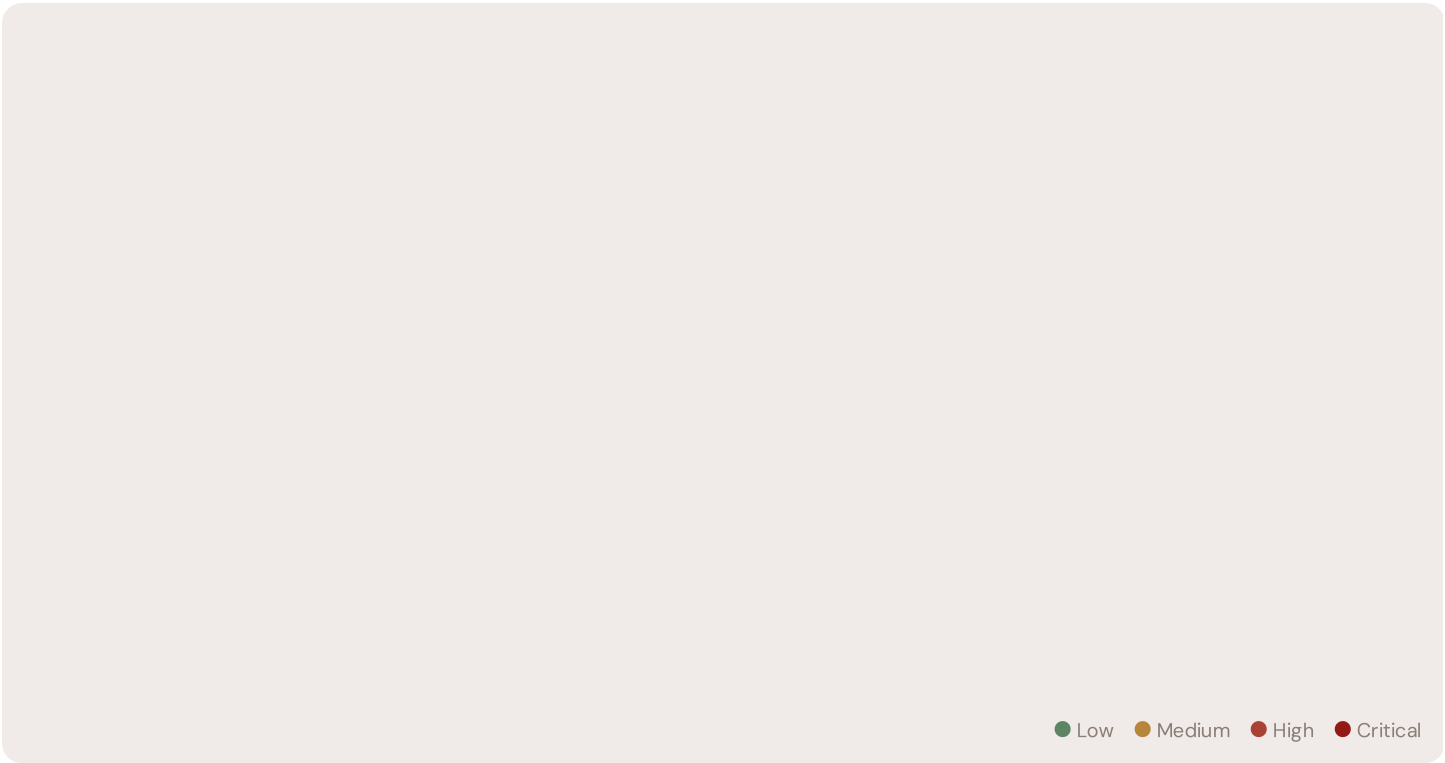
Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Dublin–Cork corridor · **002** Wind integration hotspots (Munster/Connacht) · **003** Climate vulnerability under CMIP6 · **004** Metropolitan–Rural economic impact disparity · **005** Data centre demand and grid capacity risk · **006** Infrastructure age vs smart meter rollout · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** County Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

D.1 Why the Dublin–Cork Corridor, Why Now



The Dublin hosts **346 substations** across 1 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **87 substations (25.1%)** are classified High, with only **0** in the Low band. 95% of corridor substations score above the national fleet median of 0.360. The corridor median R of 0.456 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

D.2 Corridor Map and Scoring



SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Loading...										

D.3 Component Analysis

The dominant risk driver across the Dublin corridor is **T (Transition)**, averaging 0.416 against a fleet average of 0.359 — occupying 832% of its weight ceiling ($w = 0.05$). The second contributor is **V (Voltage)** at 0.423 vs fleet 0.349 (423% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.168 (fleet: 1.095), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.028, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 County Breakdown

The region regions-level breakdown reveals significant intra-regional variation. **Leinster** leads with a mean R of 0.459 across 346 substations, while **Leinster** scores 0.459 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Dublin is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

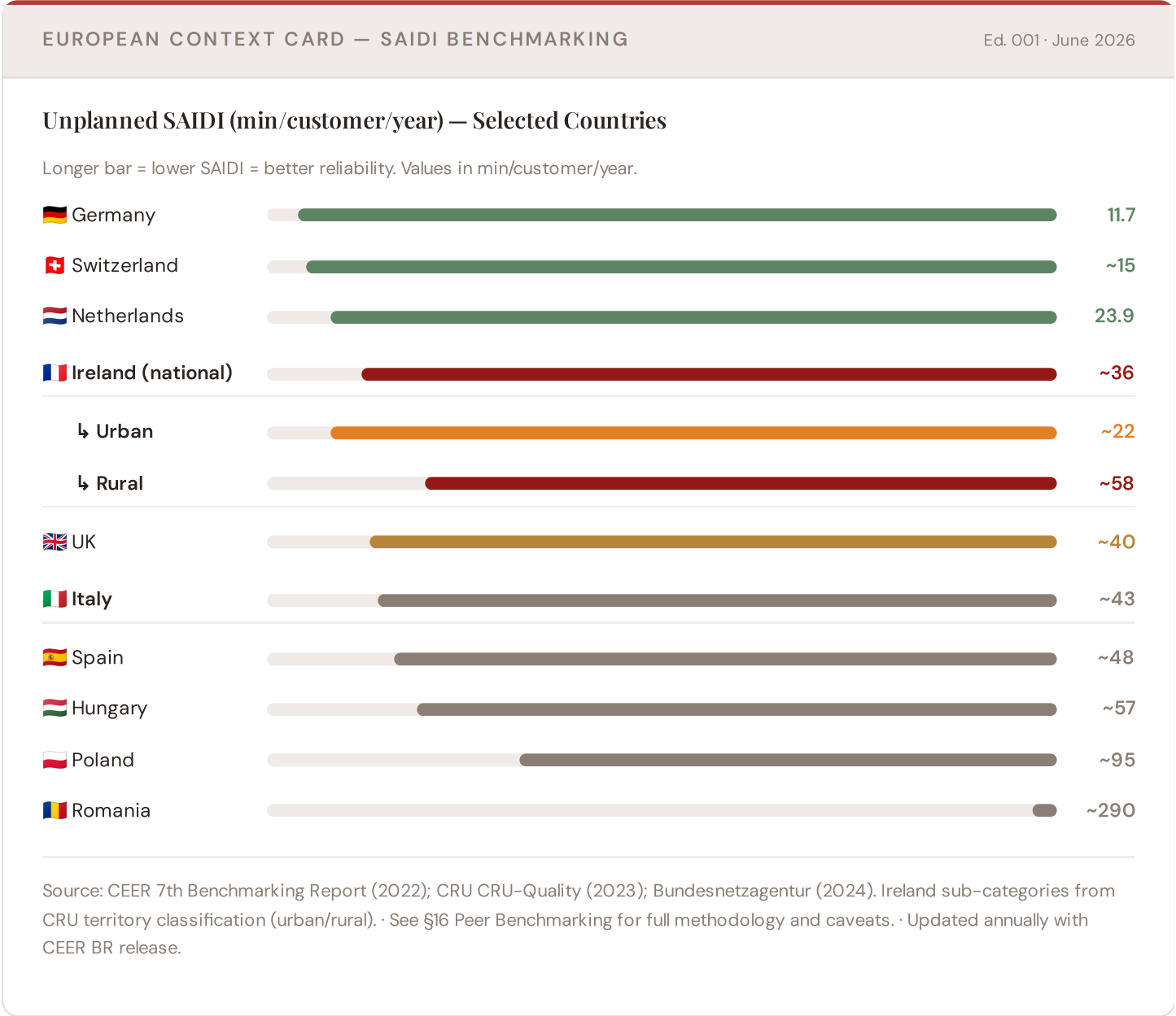
O Dublin substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Leinster sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are

disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

European Context Card

How this works: Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Dublin–Cork corridor analysis hinges on Ireland's continuity performance relative to European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.



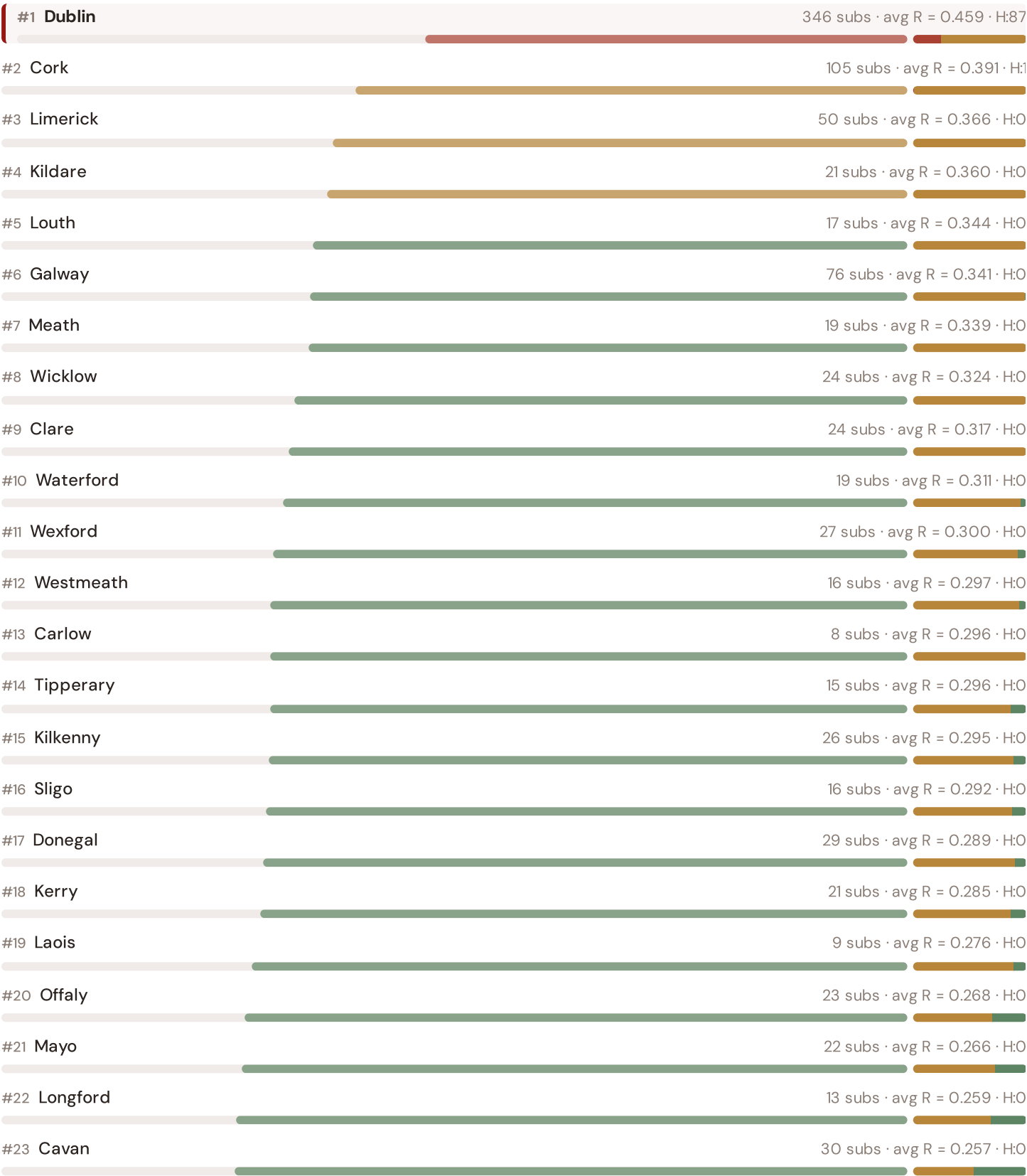
This month's key insight: The Dublin corridor deep-dive shows **346 substations** (of 346) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The

corridor's average C of 0.553 is **1.2× the fleet average** of 0.461. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regional Risk Ranking — All 26 Irish Counties

Where does the Dublin–Cork corridor sit within the national landscape? Regions sorted by mean R_{median}, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



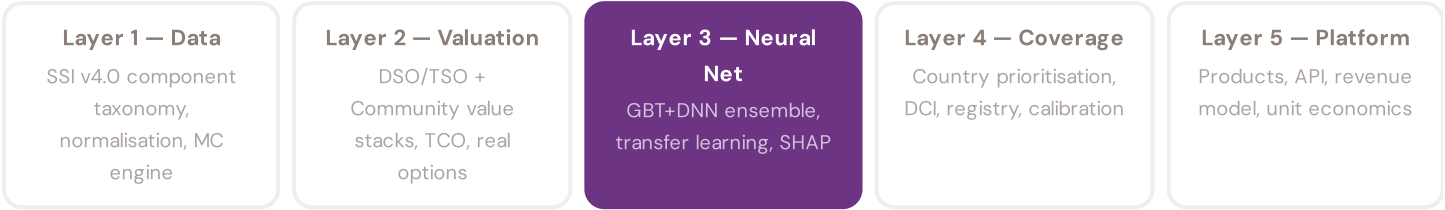


The Dublin ranks **#1 of 26 region regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are Dublin, Cork, Limerick, while the three most resilient are Leitrim, Monaghan, Roscommon. 2 of 26 score above the fleet average of 0.369. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the ESB Networks/EirGrid and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.



LAYER 3

§0.5.6 · June 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

- SHAP: game-theoretic feature attribution per prediction
- Waterfall: fleet average → substation prediction
- Direct traceability to SSI components and data sources
- Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 994 substations with complete feature vectors. Average CI_width of 0.055 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 67 Low, 839 Medium, 88 High, 0 Critical — ensures balanced training across all risk bands.

Coming next month: **LAYER 3** Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 02

Mountain Region

Deep-dive into Region 10's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Light-rural tier — limited substation density.

NEXT DATA REFRESH

Q3 2026 — 2 Quarterly Sources

SEAI, Enterprise Ireland are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **CRU (Commission for Regulation of Utilities)** (8 variables → C1–C4 (SAIDI/SAIFI), I1–I3 (IRI), quality regulation).

FLEET SPOTLIGHT

Leinster: Highest Risk Concentration

Leinster has the highest concentration of High/Critical substations: **87 of 549 (16%)** are in the upper risk bands — avg R = 0.403. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Region 10**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Irish utility, CRU, EirGrid, ESB Networks, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 003

Published 11 June 2026 · SSI Index v4.0.2 · 994 substations · 2,306 power lines · 6 components · 20 metrics · 6 modifiers
10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5